Winging It: Analyzing the Aerodynamics of Paper Planes

Namit Misra
The College of Wooster
(Dated: December 6, 2024)

I investigated the aerodynamic performance of three distinct paper plane designs— the Basic dart, the Glider, and the Phoenix. By using experimental flight data, aerodynamic parameters such as lift coefficient, drag coefficient, induced drag and zero-lift drag were analyzed. Multiple flights for each plane were recorded, and position and velocity data was extracted using high-speed video analysis software Tracker and analyzed. The Phoenix demonstrated the highest lift-to-drag ratio, of 3.63 ± 0.331. The Basic dart had the highest lift coefficient of 0.141 ± 0.0286. The Phoenix also demonstrated the lowest total drag coefficient of 0.028 ± 0.0031, and yielded the lowest induced-drag and zero-lift drag of 0.004 ± 0.001 and 0.024 ± 0.0023 respectively. For induced-drag as a ratio of total drag, the Glider was the most efficient, with the ratio being 0.08 ± 0.01.

Table I: Nomenclature

Symbol	**Description**
λ_t	Angle of sweep
ρ	Air density
AR	Aspect ratio
b	Wing span
C_D	Drag coefficient
$C_{D,i}$	Induced-drag coefficient / Drag due to lift
$C_{D,0}$	Zero lift drag coefficient
C_L	Lift coefficient
e	Oswald efficiency factor
L/D	Lift-to-drag ratio
S	Wing area
U	Relative velocity of air
W	Weight

I. INTRODUCTION

The fascination with flight has driven scientific inquiry since ancient times, leading to fundamental discoveries in aerodynamics and aircraft design. While modern aviation focuses on complex commercial and military aircraft, the study of simple flying objects remains valuable for understanding basic aerodynamic principles. Paper airplanes, in particular, serve as accessible models for investigating the relationship between structural design and aerodynamic performance.

The work of Ludwig Prandtl in the early 20th century established that aircraft efficiency is fundamentally tied to the lift-to-drag ratio, which depends critically on the structural configuration of the flying body[1]. This relationship between structure and performance remains central to aircraft design, from simple paper models to advanced aerospace vehicles. While extensive research exists on optimizing full-scale aircraft, systematic analysis of paper airplane designs has received relatively little attention in the scientific literature, despite their potential as educational and research tools.

This study investigates the flight performance of three distinct paper plane designs: the Basic dart, the Glider, and the Phoenix. Using high-speed video analysis and computational tools, key parameters are measured to analyze which design produces the best flight. The findings provide a foundation for understanding the interplay of aerodynamic forces in simplified systems.

Though focused on paper planes, the implications of this study extend to broader areas of flight and aviation. Lightweight aerial systems, such as micro aerial vehicles (MAVs) and drones, operate in regimes where small-scale aerodynamic effects dominate. These systems share challenges with paper planes, such as achieving stable flight at low speeds and optimizing performance with minimal structural support. Insights from this research can inform the design of MAVs and other unmanned systems, highlighting how the principles studied in a folded sheet of paper resonate across modern aerospace technologies.

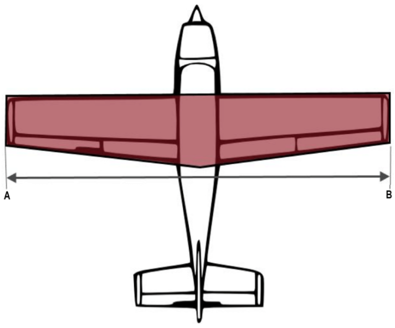

Figure 1: Wing geometry showing wingspan (b) and wing area (S) [2]

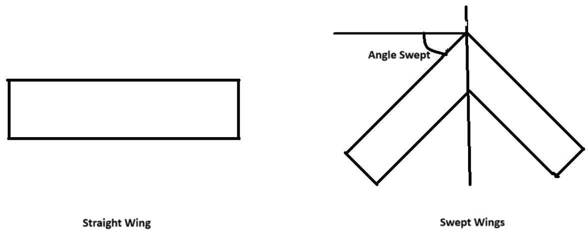

Figure 2: Wing sweep configurations

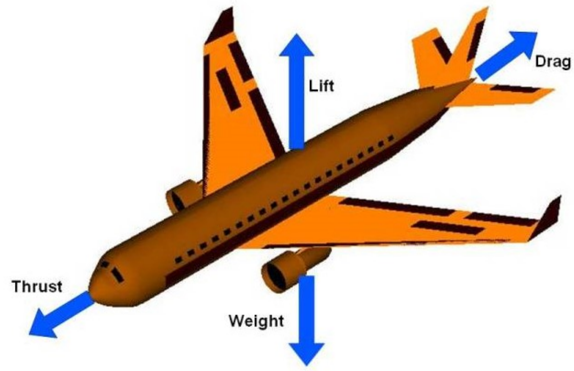

Figure 3: Forces on a aircraft [3]

II. THEORY

Though the formulas used in this study are standard aerodynamic equations designed for traditional aircraft operating at high Reynolds number i.e. turbulent fluid flow that occurs at high velocities and high altitudes, they have been adapted in various studies for use in smaller, simplified systems like paper planes, which operate under low Reynolds number conditions i.e. under laminar fluid flow[1]. The theory section is divided into two main parts. The first section focuses on the geometry of paper plane wings. These components directly influence how a plane interacts with the air and how efficient it is at generating lift while minimizing drag. Understanding the role of each component is critical for designing a plane with optimal aerodynamic properties. The second section delves into the key flight variables that determine the performance of the plane that include lift, drag, and their respective coefficients.

A. Geometry of Paper plane wing design

The design of a plane's wing is fundamental to its ability to generate lift and minimize drag. The key geometric parameters of wing design are illustrated in Fig.1. The wingspan b represents the tip-to-tip distance across the wings, while the wing area S encompasses the total surface area of the wings. The aspect ratio AR, a measure of wing slenderness, is defined as:

$$\mathrm{AR} = \frac{b^2}{S} \tag{1}$$

As shown in Fig.2, wings can be either swept or straight. The sweep angle λ_t measures angle of the wings relative to the plane's body, with straight wings having $\lambda_t = 0$.

B. Flight variables

Four forces may act upon an object in flight: lift L and weight W in the vertical direction, and thrust T and drag D in the horizontal direction, as depicted in Fig. 3 . Paper planes, lacking an engine, experience no thrust force.

The lift-to-drag ratio serves as the primary metric of aerodynamic efficiency, quantifying the amount of lift produced per unit drag[4]. For gliding flight, this ratio can be defined as

$$\frac{L}{D} = \frac{\text{Horizontal Distance travelled}}{\text{Vertical altitude lost}}. \tag{2}$$

During steady descent, lift equals weight,

$$L = W. \tag{3}$$

The lift force is defined as

$$L = \frac{1}{2} C_L \rho U^2 S \tag{4}$$

where U represents the relative velocity between the air and the plane (which equals the plane's velocity in still air), ρ is the air density, and S is the wing area [4]. The lift coefficient C_L relates the wing structure to the

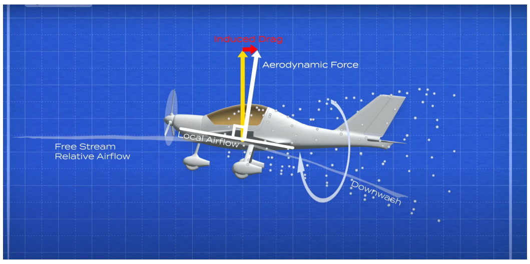

Figure 4: Animated depiction of how drag due to lift works [5].

lift produced—a higher coefficient indicates greater lift production. Using Eq. 3 and Eq. 4, the lift coefficient for a gliding plane becomes

$$C_L = \frac{2L}{\rho U^2 S}. \tag{5}$$

The drag force follows a similar form

$$D = \frac{1}{2} C_D \rho U^2 S. \tag{6}$$

where C_D represents the drag coefficient. Since Eq. 4 and Eq. 6 share the same variables except for their respective coefficients, the lift-to-drag ratio can be expressed as

$$\frac{L}{D} = \frac{C_L}{C_D}. \tag{7}$$

The drag coefficient comprises two components: induced drag ($C_{D,i}$) and drag at zero-lift ($C_{D,0}$) [4]. Induced drag arises from pressure differences between the air above and below the wing. This pressure differential creates wing-tip vortices as air swirls from beneath the plane to the top surface. As shown in Fig. 4, these vortices tilt the plane backward, reducing its lift-generating capability. The induced drag coefficient is given by,

$$C_{D,i} = \frac{C_L^2}{\pi e \mathrm{AR}}. \tag{8}$$

where e represents the Oswald efficiency factor, typically determined experimentally. While traditionally applied to full-scale aircraft, empirical formulas for estimating e can be adapted for paper planes. The formula we shall use,

$$e = \frac{2}{2 - AR + \sqrt{4 + AR^2 (1 + \tan^2 \Lambda_t)}}. \tag{9}$$

Thus far, we have established that drag comprises two key components: induced drag and zero-lift drag. Induced drag arises from the wing's lift-generating mechanism and determines the efficiency of a plane's lift generation. Looking at Eq. 8, we can see that the induced drag coefficient is proportional to the square of the lift coefficient—higher lift inevitably leads to greater induced drag. However, this drag penalty can be minimized through efficient wing design, as represented by the denominator terms πeAR. To better understand the impact of induced drag, we can examine its ratio to total drag, which quantifies the amount of total drag due to induced drag. Hence, a high induced drag ratio would mean that the structure of the plane produces lower drag at near-zero lifts, but fails to do so at higher lifts due an inefficient wing structure.

Zero-lift drag, which depends on the entire plane structure (the body and the wings) and is experienced by the plane even if it generates no-lift, such as during landing. This can be expressed as,

$$C_{D,0} = C_D - C_{D,i}. \tag{10}$$

Note that L/D, C_L, C_D, $C_{D,i}$, and $C_{D,0}$ are all dimensionless quantities. The optimal aerodynamic design strikes a delicate balance: it must maximize lift while minimizing both induced drag and zero-lift drag components. This balance determines the plane's overall performance and efficiency in flight, with the most successful designs achieving high lift-to-drag ratios through careful optimization of these interconnected parameters.

III. PROCEDURE

A. Setup

Three paper plane designs were selected for the experiment: the Basic dart[6], the Glider[7], and the Phoenix. Each plane was constructed from standard letter sized construction paper (8.5 inches by 11 inches) to maintain consistency in material and size. Two planes were constructed for each design, although only one was used throughout the trials. After construction, key physical properties of each plane were recorded, including the wingspan, wing area, and weight A bar was clamped to a table and was used to mark the launch height for every flight to ensure uniformity. A GoPro Hero6 was setup on a tripod approximately 3 meters away from the launch point, in the center of the room where the data was being recorded. The videos were recorded at 120 frames per second.

B. Data Collection

Each plane was launched multiple times to obtain five usable recordings for the Glider and the Phoenix, where the entire flight remained in the field of view and the trajectory was relatively straight. The Basic dart's poor flight performance required multiple launches to achieve successful flights, limiting data collection to

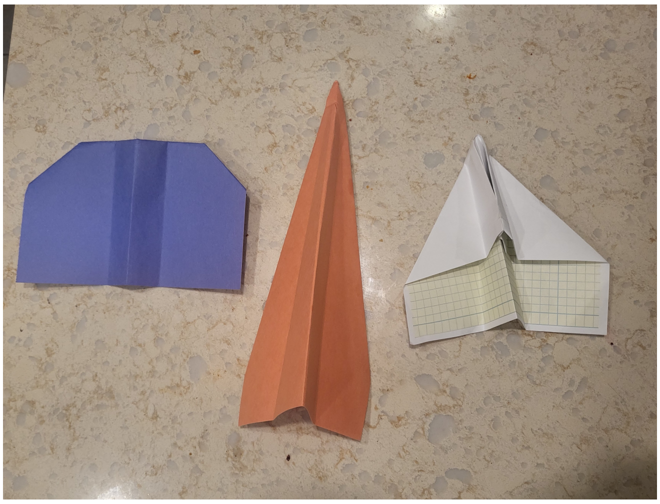

Figure 5: Different airplane design. From the left to right: the Glider (in blue), the Basic dart (orange), and the Phoenix (white)

three recordings.
The videos of the flights were exported to Tracker, which is a video analysis and modeling tool. For scaling, the length of a whiteboard that was placed behind the planes was measured.The coordinate axes were defined with the plane's initial position as origin. The Autotracker option was used to track the position of the of the tail of planes, with manual marking done wherever necessary. Then, Tracker would use the markers and the scaling to measure the x and y positions of the planes in each frame. Tracker also extrapolated the velocity of the plane in each frame using the measured positions and time between every frame. Velocity and position data was exported to Igor for analysis.

IV. Preface to Analysis: Obtaining the Required Coefficients

Table.II lists all the geometrical parameters of the wings for all three planes. These value shall be used throughout the analysis.

To fully understand the aerodynamic performance of the paper planes, a series of calculations were performed using experimental data extracted from high-speed video analysis. Through an example using the data collected in trial 1 for the Basic dart, this section details the methodology used to compute the key aerodynamic parameters, providing a clear pathway from raw data to meaningful metrics. While the formulas employed are traditionally used for conventional aircraft, their application to paper planes offers a simplified yet insightful perspective into the fundamental principles of flight.

For a the first trial of the Basic dart, the following measurements were obtained using Tracker software:

- Horizontal distance traveled: 5.860 m
- Vertical descent: 2.933 m
- Average velocity (U): 9.74 m/s
- Wing area (S): 0.0097 m²
- Air density (ρ): 1.225 kg/m³

Starting with the lift-to-drag ratio, using Eq. 2 we obtain,

$$\frac{L}{D} = \frac{\text{Horizontal distance}}{\text{Vertical descent}} = \frac{5.860 \text{ m}}{2.933 \text{ m}} = 1.998. \tag{11}$$

For the lift coefficient calculation, we use Eq. 5. Under steady gliding conditions, lift equals weight, hence

$$\begin{aligned} C_L &= \frac{2L}{\rho U^2 S} \\ &= \frac{2(0.0579 \text{ N})}{(1.225 \text{ kg/m}^3)(9.74 \text{ m/s})^2(0.0097 \text{ m}^2)} = 0.104 \end{aligned} \tag{12}$$

The drag coefficient can then be determined using Eq. 7,

$$C_D = \frac{C_L}{L/D} = \frac{0.104}{1.998} = 0.052. \tag{13}$$

To find the induced ratio, we use Eq.8. Substituting the known values,

$$C_{D,i} = \frac{C_L^2}{\pi e \text{AR}} = \frac{(0.104)^2}{\pi(0.89)(2.42)} = 0.0063. \tag{14}$$

The induced drag coefficient was simply divided by the drag coefficient to calculate the ratio. Finally, the zero-lift drag coefficient is calculated using Eq. 10,

$$C_{D,0} = C_D - C_{D,i} = 0.0517 - 0.0063 = 0.0453 \tag{15}$$

These calculations were repeated for each flight. Table.III depicts these values in a tabular form.

V. Results and Analysis

A. Lift to drag ratio

Fig.6 depicts the plot of lift-to-drag ratio of different plane types across all trials. The Phoenix had the best aerodynamic performance, with the highest mean lift-to-drag ratio of 3.63 ± 0.332, followed by the Basic dart with a value of 2.53 ± 0.124, and the Glider which yielded a lift-to drag ratio of 2.17 ± 0.193.

Table II: Geometrical parameters of the wings for all three plane designs

Plane Type	Wing area	Aspect ratio	Angle swept	Oswald efficiency factor e
Glider	$0.020 \pm 0.001 m^2$	1.6 ± 0.2	$40° \pm 1°$	0.61
Basic dart	$0.0097 \pm 0.001 m^2$	1.4 ± 0.2	$75° \pm 1°$	0.39
Phoenix	$0.016 \pm 0.001 m^2$	1.2 ± 0.2	$45° \pm 1°$	0.57

Table III: Calculated coefficient values for example flight

	Lift-to-Drag ratio	C_L	C_D	$C_{D,i}$	$C_{D,0}$
Example Fight	1.998	0.104	0.0517	0.063	0.0453

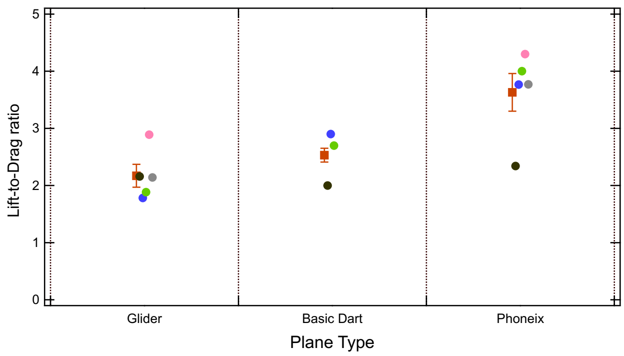

Figure 6: Plot of lift-to-drag ratio vs Plane type. Individual trials shown in different colors: trial 1 (black), trial 2 (blue), trial 3 (green), trial 4 (pink), and trial 5 (grey). Red square markers indicate average values, which were as follows: 2.17 ± 0.193 for the Glider, 2.53 ± 0.124 for the Basic dart, 3.63 ± 0.331 for the Phoenix.

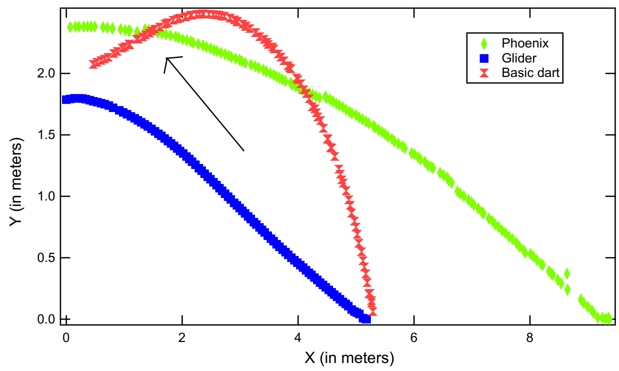

Figure 8: y vs x position graph for all three paper planes. Green points depict the Phoenix, blue markers depict the Glider, and red markers depict the Basic dart. The arrow points highlights the extra lift produced by the Basic dart, compared to the other two that had a steady descent

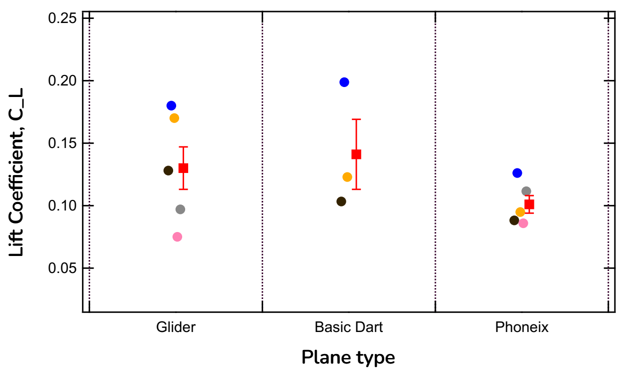

Figure 7: Plot of lift coefficient (C_L) vs Plane type. Trial colors as described in Fig. 6. Average values: 0.13 ± 0.02 for the Glider, 0.141 ± 0.028 for the Basic dart, and 0.101 ± 0.007 for the Phoenix.

B. Coefficient of Lift

Fig.7 depicts the plot of lift coefficient, C_L with respect to the plane type across all trials. The Phoenix exhibited the lowest mean lift coefficient of 0.101 ± 0.007, despite achieving the highest lift-to-drag ratio indicating that its superior performance stems from factors other than its lifting capability. Basic dart had the highest lift coefficient, C_L of 0.141 ± 0.0286, followed by the Glider with the value 0.13 ± 0.02. These values reveal that higher lift generation capability does not necessarily translate to best aerodynamic performance.

An interesting observation was made for the flight of the Basic dart plane. When launched, while the other planes showed a steady descent, the Basic dart had an additional lift before it crashed to the ground. This is depicted in a y vs. x position plot in Fig.8. As the arrow highlights, the Basic dart moves higher, generating more lift before descending. All these flights were at similar angles hence an angle dependence can be ruled out. Further investigation into this is required.

C. Drag Coefficient

The plot of drag coefficient with respect to plane type is presented in Fig.9. The Phoenix demonstrated remarkably low drag with a mean coefficient of 0.028 ± 0.003, which explains its high lift-to-drag ratio despite producing low lift. The Basic dart and the Glider had

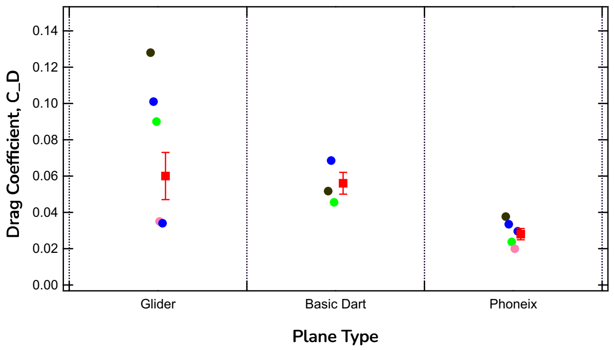

Figure 9: Plot of drag coefficient, C_D vs Plane type. Trial colors as described in Fig. 6. Average drag coefficient, C_D , was 0.06 ± 0.01 for the Glider, 0.056 ± 0.006 for the Basic dart, and 0.028 ± 0.003 for the Phoenix.

very similar drag coefficients with a value of 0.056 ± 0.006 and 0.06 ± 0.01 respectively.

D. Induced drag coefficient

Induced drag coefficient represents the drag produced in a paper plane due to its lift. Plot. 1 in Fig.10 represents a plot of induced drag coefficient, $C_{D,i}$ with respect to plane type. The Phoenix had the lowest induced drag coefficient of 0.004 ± 0.0007, followed by the Glider with a value of 0.006 ± 0.001, and the Basic dart at 0.012 ± 0.005.
To provide meaningful context for these values, we calculated the ratio of induced drag to total drag coefficient. This ratio reveals what fraction of each plane's total drag results from lift generation. A lower induced drag ratio suggests that most of the drag comes from other sources and would act upon the plane even at zero lift, while a higher ratio indicates that lift production is the dominant source of drag, and the plane would be unable to perform well even if it were to produce higher lift.

Plot 2. in Fig.10 depicts a plot of induced drag coefficient as a ratio of total drag coefficient. The Glider exhibited the lowest induced-to-total drag ratio at 0.08 ± 0.01, indicating that only 8% of its total drag comes from lift generation. This reveals that while the Phoenix had the lowest absolute induced drag, the Glider more effectively minimizes the drag associated with lift production relative to its total drag. Hence, most of the Glider's drag stems from other sources and should be apparent when zero-lift drag coefficient is calculated in the following section.
The Phoenix's low induced drag coefficient of 0.004 ± 0.0007 highlights its ability to generate lift efficiently.

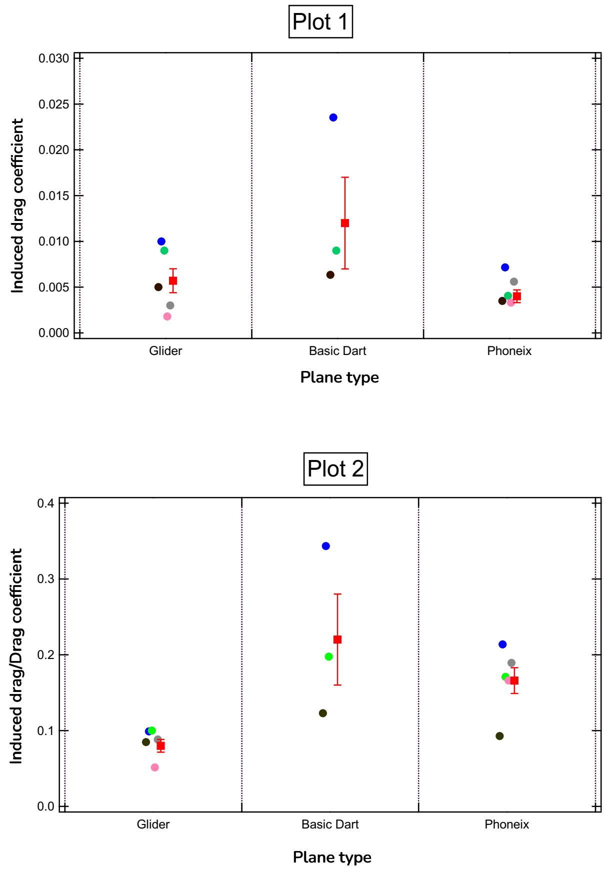

Figure 10: Plot 1: Induced drag coefficient ($C_{D,i}$) vs. Plane type and Plot 2: Ratio of induced to total drag ($C_{D,i}/C_D$) vs Plane type. The Glider, Basic dart, and Phoenix showed mean $C_{D,i}$ values of 0.006 ± 0.001, 0.012 ± 0.005, and 0.004 ± 0.0007 respectively, with corresponding $C_{D,i}/C_D$ ratios of 0.08 ± 0.01, 0.22 ± 0.06, and 0.17 ± 0.02.

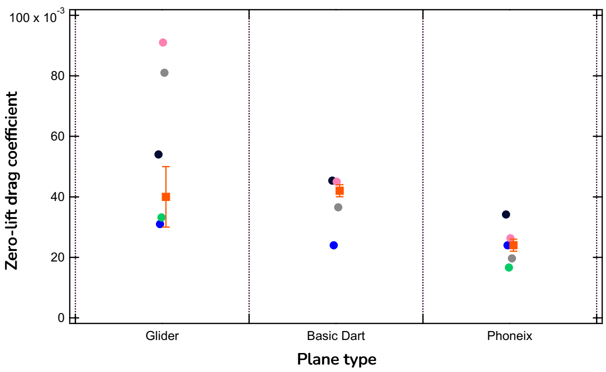

Figure 11: Plot of zero-lift drag coefficient $C_{D,0}/C_D$ vs Plane type. rial colors as described in Fig. 6. The average value was 0.04 ± 0.01 for the Glider, 0.042 ± 0.002 for the Basic dart, and 0.024 ± 0.0023 for the Phoenix.

E. Zero-lift drag coefficient

Zero-lift drag coefficient represents the drag a plane produces at zero-lift. The Phoenix had the lowest value, with $C_{D,0}$ being 0.024 ± 0.002, followed by the Basic dart and Glider with similar values of 0.042 ± 0.0023 and 0.04 ± 0.01 respecitvely. This is depicted in Fig.11. These results indicate that the Phoenix's design fundamentally minimizes drag through its structure, contributing to its overall superior aerodynamic performance.

VI. CONCLUSION

This study explored the aerodynamic performance of three distinct paper plane designs—Basic dart, Glider, and Phoenix—by analyzing their lift-to-drag ratios, lift and drag coefficients.

The Phoenix exhibited the best overall aerodynamic performance, with the highest mean lift-to-drag ratio of 3.63 ± 0.332 and the lowest drag coefficient of 0.028 ± 0.0032. This superior efficiency was primarily attributed to its low drag, despite having the lowest lift coefficient of 0.101 ± 0.007. In contrast, the Basic dart displayed unique flight characteristics, including an additional lift observed during its trajectory. While its lift coefficient 0.141 ± 0.0286 was the highest among the three planes, it also had an inefficient structure as suggested by its induced drag and zero-lift drag coefficients of 0.012 ± 0.005 and 0.042 ± 0.0023 respectively.

The Glider, characterized by its large wing area and high aspect ratio, showed steady descent and the least induced drag ratio of 0.08 ± 0.01 relative to its total drag.

These findings underscore the importance of understanding aerodynamic principles in unconventional contexts like paper planes. By bridging experimental data with aerodynamic theory, this study offers valuable insights into how design and launch conditions affect flight. Beyond paper planes, these results have broader implications for low Reynolds number flight, such as in micro aerial vehicles (MAVs) and lightweight drones, where balancing lift and drag is essential for efficient performance.

VII. FUTURE WORK

Future estudies could focus on analyzing the trends observed in this experiment to better understand the specific design components that influence aerodynamic performance. For instance, examining how variables such as aspect ratio, wing shape, and weight distribution contribute to lift-to-drag efficiency could provide deeper insights into the observed differences among the plane designs. Additionally, further investigation into the angle dependence of flight performance could help identify optimal launch conditions. The results from this study can be further confirmed using a wind tunnel setup. These explorations would not only enhance the understanding of paper plane dynamics but also have broader applications in low Reynolds number aerodynamics.

VIII. ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Dr. Susan Lehman for her invaluable guidance and support throughout the course of this experiment. I would also like to thank Mr. Royse for his assistance during the data collection phase. His help in setting up the equipment and ensuring smooth data recording made the experiment run efficiently.

[1] D. Raymer, *Aircraft design: a conceptual approach* (American Institute of Aeronautics and Astronautics, Inc., 2012).

[2] ResearchGate, *A wingspan is the width of the wings projection on the horizontal plane from tip to*, https://www.researchgate.net/figure/A-wingspan-is-the-width-of-the-wings-projection-on-the-horizontal-plane-from-tip-to$_{f}ig11_{3}45268053(2024), accessed : November 29, 2024$.

[3] NASA Glenn Research Center, *Beginner's guide to aerodynamics*, https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/learn-about-aerodynamics/ (2024), accessed: November 29, 2024.

[4] J. D. Anderson and M. L. Bowden, *Introduction to flight*, vol. 582 (McGraw-Hill Higher Education New York, NY, USA, 2005).

[5] Flight Club, *What is induced drag*, https://youtu.be/MnB6Lqr91Yc (2022), youTube video, accessed: November 29, 2024.

[6] F. N. Fly, *Tutorial on how to fold a dart paper plane* (n.d.), accessed: 2024-12-06, URL https://www.foldnfly.com/1.html.

[7] F. Flight, *Tutorial on how to fold glider* (n.d.), accessed: 2024-12-06, URL https://www.youtube.com/watch?v=QOGCD4_pqdA.